Quantitative Colour Pattern Analysis (QCPA)

Post-batch processing and analysis in R

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1 Setup

The following R script uses the trichromatic example without UV (sea slugs seen by a triggerfish, *Rhinecanthus aculeatus*), but can similarly be applied to the tetrachromatic example with UV (spiders seen by a bluetit, *Parus major*). To make the worked example readily accessible to users, we have prioritised using base R functions. As we will be working with primarily character-type data (i.e., species names, region of interest descriptors), we will introduce the user to many base functions that are for list-type data structures (e.g., `lapply()`). For clarity, we recommend that you, as the user, need to be comfortable with the different data types (e.g., numerical, character, factors) and data structures such as lists (collection of mixed data types) and data frames (special type of list). We also provide different methods to extract data, ideally introducing different coding methods and providing ways to increase the efficiency of data wrangling.

For the following R script to run, please ensure that you have implemented the QPCA batch script methods. In the following folder

`Worked example data/Example1_tri-chromat_no_UV/Test Data`

there should be two subfolders for the two species, *Aphelodoris varia* and *Aplysia sp.* Within these subdirectories, please confirm that data has been generated. As you proceed through the next few steps, we provide code that checks the data totality. If you are uncertain that you uniformly named files, we have additionally provided script to check region of interest names (see section 2.3).

2 Create the foundational dataset identifying unique animal IDs and regions of interest for multiple viewing distances

2.1 Iterate over folders to extract species, animals and viewing distances

First, we will specify the location of the data files with respect to where your working directory has been saved. Here, we have assumed that the R script was saved in the parent folder of Worked example data. The next step will be to extract the names of the two focal species using the folder names within the `Test Data` folder.

```
# Get directory; assumes the R script is in the parent folder of Worked example data
pth <- getwd()

# Set sub folder name containing your data
Extrct_Fold <- "Worked example data/Example1_tri-chromat_no_UV/Test Data"

# Paste the two locations together to specify your global data location
loc <- paste(pth, Extrct_Fold, sep = "/")

# Detect species -these could also be locations depending on your data set.
Species <- list.files(path = loc)

Species # Should contain species folders and a log file with QCPA processing details
```

```
## [1] "Aphelodoris varia"  "Aplysia sp"         "Batch_QCPA_Log.txt"
```

```
# Here, we use the "Ap" pattern in both species names to subset the Species list
Species <- Species[grepl("Ap", Species)] # Removes the log file from vector

# Note: For multiple different species patterns, can search using the or condition '|'
```

```r
# e.g., Species<-Species[grepl("A|B", Species)], returns species with pattern 'A' or 'B'

Species
```

```
## [1] "Aphelodoris varia" "Aplysia sp"
```

Here, we will demonstrate extracting folder information names using a for loop. We first create an empty data frame called `Animal_Info`. Then, for each ith species (here $i = 1$ or 2), we create a path into the species folder and record in the object `Indviduals_list` all the unique individual animal folder names using the `list.files` function. We then establish a second, nested loop, which will then iterate over each individual (`ind`) in `Indviduals_list` to extract all unique viewing distance folder names using `list.files`. To distinguish the viewing distance folders from other files saved within the directory, use the argument `pattern = "cm"` in `list.files`. Finally, we combine the unique species name, individual name and viewing distances into unique columns and add as a new row in `Animal_Info`.

```r
Animal_Info <- data.frame(NULL)  # First create an empty data frame

# Then create a loop to iterate over the species folders,
# extracting viewing distances and individual animal IDs
for (i in 1:length(Species)) { # Here i will be either 1 or 2
  Ind_pth <- paste0(loc, "/", Species[i]) # Pastes the ith species from the Species object
  Indviduals_list <- list.files(path = Ind_pth) # Extracts all individual folder names

  # Iterate over the number of individuals in each species folder
  for (ind in 1:length(Indviduals_list)) {
    # The variable dist lists all the unique viewing distance folder names
    # identified by "cm"
    Dists <- list.files(paste0(Ind_pth, "/", Indviduals_list[ind]), pattern = "cm")

    # Add to Animal_Info a new row with species name, individual and viewing distance
    rbind(Animal_Info, cbind(Species[i], Indviduals_list[ind], c(Dists))) -> Animal_Info
  }
}
colnames(Animal_Info) <- c("Species", "Ind", "Dist") # Rename the column names in Animal_Info
```

2.2 Inspect data

Check that the data has correctly been imported and reflects your experimental records. We converted the character data into factors, which is the appropriate format for running statistical tests in downstream analyses. We use the `lapply()` function to apply the function `factor()` to each column in our data frame. We can again use the `lapply()` function to apply the function `nlevels()` to our newly converted data frame, which will report the number of unique elements within each column of the data frame `Animal_Info`.

```r
# Check data
Animal_Info
```

```
##             Species                            Ind Dist
## 1 Aphelodoris varia 3_AphelodorisVaria_10_NF_D_RAW 30cm
## 2 Aphelodoris varia 3_AphelodorisVaria_10_NF_D_RAW 50cm
## 3 Aphelodoris varia 3_AphelodorisVaria_11_NF_D_RAW 30cm
## 4 Aphelodoris varia 3_AphelodorisVaria_11_NF_D_RAW 50cm
```

```
## 5        Aplysia sp         4_Aplysiasp_1_NL_N_RAW 30cm
## 6        Aplysia sp         4_Aplysiasp_1_NL_N_RAW 50cm
```

```r
# Convert character data into factors
Animal_Info[, colnames(Animal_Info)] <- lapply(Animal_Info[, colnames(Animal_Info)], factor)

str(Animal_Info) # Observe structure of the data
```

```
## 'data.frame':    6 obs. of  3 variables:
##  $ Species: Factor w/ 2 levels "Aphelodoris varia",..: 1 1 1 1 2 2
##  $ Ind    : Factor w/ 3 levels "3_AphelodorisVaria_10_NF_D_RAW",..: 1 1 2 2 3 3
##  $ Dist   : Factor w/ 2 levels "30cm","50cm": 1 2 1 2 1 2
```

```r
# Check the number of unique species, individuals and viewing distances
lapply(Animal_Info, nlevels)
```

```
## $Species
## [1] 2
##
## $Ind
## [1] 3
##
## $Dist
## [1] 2
```

```r
# To visualise the experimental design, we first aggregate the data to determine the
# number (length) of each individual by species and distance
Ani_info_tab <- aggregate(Ind ~ Species + Dist, data = Animal_Info, length)
barplot(Ind ~ Dist + Species,
  beside = TRUE, ylab = "Count",
  legend.text = TRUE, data = Ani_info_tab
)
```

In this example, there are only three individuals. However, with a larger data set with missing data, this becomes an important step to determine a balanced experimental design.

2.3 Inspect regions of interest for inconsistent labelling (additional step)

The number of data files will depend on whether animal, background and/or animal+background options were selected; this will make up the prefix of the file name followed by an underscore and image analysis file type (e.g., '*_Summary Results.csv*'). For the following R code to work, regions of interest must be uniformly named and capitalised for all individuals analysed. Below is a method to ensure that these regions of interest (ROIs) have been correctly named by reporting all unique names across your data by examining all unique `.tif` file names. If you are confident that the naming was consistent, you may skip this step.

```r
# Create vector containing all the individual animal sub directories
tmpAnidir <- apply(Animal_Info, 1, function(x) paste(unlist(x), collapse = "/"))

# Paste the global file directories to the front of the sub directory paths
tmpAnidir <- paste(loc, tmpAnidir, sep = "/")
```

Figure 1: Figure 1. Counts of individuals within species for each viewing distance.

```r
# This should highlight the unique regions of interest, if there are
# more than expected - look for naming inconsistencies
unique(unlist(strsplit(list.files(tmpAnidir, pattern = ".tif"), "_"))[c(TRUE, FALSE)])
```

```
## [1] "animal"            "animal+background" "background"
```

How the nested line works

The above line can be broken down into four steps:

1. `list.files()` searches all the locations specified in the tmpAnidir vector for files with the `.tif` extension and puts these file names into a new character vector.
2. Knowing that file names are composed of two bits of information separated by an underscore, `strsplit()` cuts the filenames into two elements at the underscore and outputs a list containing these two new elements for each unique file.
3. `unlist()` puts all elements into a vector. As the region of interest are the prefix, it will be first element in the pair, which we can subset using: `[ c(TRUE,FALSE)]`.
4. Finally, the `unique()` function looks across all names used for each individual and retains only unique names, omitting duplicates.

2.4 Add regions of interest to Animal information dataset

To proceed with the next step, the regions of interest (ROIs) must be uniformly named across all individual specimens (see above if uncertain). Here, we introduce two methods to add ROIs. The first method is the quickest, and works if you are confident in your naming conventions.

However, we provide a second method that extracts your data's ROIs.

2.4.1 Simplified method to list ROIs

```r
ROI <- c("animal", "animal+background", "background") # Add the names of ROIs
```

2.4.2 Extract ROIs from the data

Here, we will utilise the batch script naming convention of using the ROI as the prefix of the image analysis file name followed by an underscore and image analysis file type (e.g., '*_Summary Results.csv*'). See Table 1 below for all QCPA analyses and associated file names.

Table 1: Table 1. QCPA image analysis and associated output files. See van den Berg & Troscianko et al. (2020) for a detailed description.

Image analysis	File name
CAA, VCA, BSA	_Summary Results.csv
GabRat	_GabRat_Results.csv
LEIA	_Local Edge Intensity Analysis.csv
Particle analysis	_Cluster Particle Analysis Summary Results.csv
Particle analysis + cluster properties	_Cluster Results.csv
Particle analysis + sub-cluster properties	_Individual Particle Results.csv
Particle analysis + transition matrix + cluster properties	_ROI Cluster Results.csv

We can then extract any particular image analysis .csv data files names using the `list.files()` function specifying the file name as the pattern to search for. In the following code, we have used 'Summary Results.csv*' as our string pattern. Once we have all the file names with 'Summary Results.csv*', we remove this phrase using `gsub()`.

```
# Here, we create a character vector determining whether animal,
# background and/or animal+background options were used
# Using the first row of Animal_Info to specify the location to examine
# the number and types of _Summary Results.csv files

# Find all the ROI output files for '_Summary Results.csv'
ROI <- list.files(paste(loc, paste(unlist(Animal_Info[1, ]), collapse = "/"),
  sep = "/"
), pattern = "_Summary Results.csv")

# Here we remove the "_Summary Results.csv" information
ROI <- gsub("_Summary Results.csv", "", ROI)
ROI # In the QCPA batch script, we have investigated three ROIs
```

```
## [1] "animal"            "animal+background" "background"
```

2.5 Merge the list of ROIs names with the animal ID dataframe

```
# Expand each unique row of the Animal_Info data frame to included a unique ROI value
Animal_Info_ROI <- merge(Animal_Info, ROI)

colnames(Animal_Info_ROI)[4] <- "ROI"  # Name column 4 to ROI

Animal_Info_ROI$ROI <- factor(Animal_Info_ROI$ROI)  # Convert to factor

head(Animal_Info_ROI)  # Check data
```

```
##             Species                            Ind Dist    ROI
## 1 Aphelodoris varia 3_AphelodorisVaria_10_NF_D_RAW 30cm animal
## 2 Aphelodoris varia 3_AphelodorisVaria_10_NF_D_RAW 50cm animal
## 3 Aphelodoris varia 3_AphelodorisVaria_11_NF_D_RAW 30cm animal
```

```
## 4 Aphelodoris varia 3_AphelodorisVaria_11_NF_D_RAW 50cm animal
## 5        Aplysia sp         4_Aplysiasp_1_NL_N_RAW 30cm animal
## 6        Aplysia sp         4_Aplysiasp_1_NL_N_RAW 50cm animal
```

```r
str(Animal_Info_ROI)  # Check that ROI is a factor
```

```
## 'data.frame':    18 obs. of  4 variables:
##  $ Species: Factor w/ 2 levels "Aphelodoris varia",..: 1 1 1 1 2 2 1 1 1 1 ...
##  $ Ind    : Factor w/ 3 levels "3_AphelodorisVaria_10_NF_D_RAW",..: 1 1 2 2 3 3 1 1 2 2 ...
##  $ Dist   : Factor w/ 2 levels "30cm","50cm": 1 2 1 2 1 2 1 2 1 2 ...
##  $ ROI    : Factor w/ 3 levels "animal","animal+background",..: 1 1 1 1 1 1 2 2 2 2 ...
```

3 Create a function to import a .csv list from VCA,BSA,CAA data, with options for particle analysis

As before, you could use a series of nested loops to extract the data in the csv files (e.g., section 2.1). However, here we introduce how to define a function to determine the data location using the data listed in the *Animal_Info_ROI* data frame and import the .csv file. By creating a function, we can then make our file importing dynamic, providing options to import different image analysis files. And, by using information called directly from the *Animal_Info_ROI* data frame, we ensure the data extracted is correct and reduces errors that would have potentially occurred if we did a broad import. First, we present a simple example of a function to import the VCA,BSA,CAA data files, and then, secondly, we expand on our function to demonstrate how to define and import multiple types of image analysis data files.

3.1 Simplified function example to read in image analysis files

In the following function, we walk through the steps for a function that only imports VCA, BSA, CAA image analysis data but will lay the foundations for when we create the function that can input any of the image analyses.

There are three parts:

1. The first section will specify what type of image analysis we are interested in
2. The second section extracts the data using `Animal_Info_ROI` data frame and stores the data as a list in the object `Sum_Res_list`
3. In the third section, we use the `do.call()` function to extract all the list elements in `Sum_Res_list` and combine them into a data frame using `rbind()`. From Table 1, the file suffix for this analysis is '*_Summary Results.csv*'.

```r
# Part A:
# Here, we specify what image analysis so that we can use the correct file extension
exnfile <- "_Summary Results.csv"

Ani_ID_dat <- Animal_Info_ROI # Assign our animal ID data frame

path <- loc # This is the global location set in section 2.1

# Part B:
# Once the file extension condition is met, we use lapply to extract the data,
# using a nested function within lapply
```

```r
Sum_Res_list <- lapply(
  1:nrow(Ani_ID_dat), # for each row in Ani_ID_dat...
  function(x) { # start of our nested function

    # Extract the individual row information
    tmpLoc <- paste(unlist(Ani_ID_dat[x, ]), collapse = "/")

    # Create a directory string
    tmpLoc <- paste0(paste(path, tmpLoc, sep = "/"), exnfile)

    # Read in the data using the newly specified location
    data <- read.csv(tmpLoc)

    # Add the specific animal information to the data
    data <- merge(Ani_ID_dat[x, ], data)

    data # Returns data
  }
)

# Part C:
# We then collapse the Sum_Res_list list into a data frame by
# calling rbind in the function do.call
Sum_Res_df <- do.call(rbind, Sum_Res_list)

# Check the imported data
dim(Sum_Res_df) # 18 rows by 133 columns
```

```
## [1]  18 133
```

```r
head(Sum_Res_df)[, 1:12] # check the first six rows and first twelve columns
```

```
##             Species                            Ind Dist    ROI X
## 1 Aphelodoris varia 3_AphelodorisVaria_10_NF_D_RAW 30cm animal 1
## 2 Aphelodoris varia 3_AphelodorisVaria_10_NF_D_RAW 50cm animal 1
## 3 Aphelodoris varia 3_AphelodorisVaria_11_NF_D_RAW 30cm animal 1
## 4 Aphelodoris varia 3_AphelodorisVaria_11_NF_D_RAW 50cm animal 1
## 5        Aplysia sp         4_Aplysiasp_1_NL_N_RAW 30cm animal 1
## 6        Aplysia sp         4_Aplysiasp_1_NL_N_RAW 50cm animal 1
##                                                                                    Image
## 1 3_AphelodorisVaria_10_NF_D_RAW.ORF_Acuity3_Dist300_0_WeberFiltered_itr5_rad5_fall3_0
## 2 3_AphelodorisVaria_10_NF_D_RAW.ORF_Acuity3_Dist500_0_WeberFiltered_itr5_rad5_fall3_0
## 3 3_AphelodorisVaria_11_NF_D_RAW.ORF_Acuity3_Dist300_0_WeberFiltered_itr5_rad5_fall3_0
## 4 3_AphelodorisVaria_11_NF_D_RAW.ORF_Acuity3_Dist500_0_WeberFiltered_itr5_rad5_fall3_0
## 5    4_AplysiaParvula_1_NL_N_RAW.ORF_Acuity3_Dist300_0_WeberFiltered_itr5_rad5_fall3_0
## 6    4_AplysiaParvula_1_NL_N_RAW.ORF_Acuity3_Dist500_0_WeberFiltered_itr5_rad5_fall3_0
##     CAA.Sc    CAA.Jc   CAA.St    CAA.Jt     CAA.Hc     CAA.Qc
## 1 1.000000 1.0000000      Inf       NaN 0.00000000        NaN
## 2 1.000000 1.0000000      Inf       NaN 0.00000000        NaN
## 3 3.407769 0.8519422 3.086572 0.5144287 1.29367056 0.93318605
## 4 3.321784 0.8304460 2.706124 0.4510207 1.28003640 0.92335109
## 5 2.095732 0.6985775 1.219512 0.4065041 0.80674082 0.73432714
## 6 1.020302 0.5101512 1.000000 1.0000000 0.05623232 0.08112609
```

```r
# Finally, to prevent any errors we remove the objects Ani_ID_dat, exnfile, path
# from the global environment because we'll be recycling them in the second function
rm(exnfile)
rm(Ani_ID_dat)
rm(path)
```

3.2 Dynamic function to read in various image analysis files

To dynamically import data, we are going to create the function `readQCPA()`, which will take an animal ID data frame (`Ani_ID_dat`), the specific image analysis that we are interested in (`filetype`), and a folder directory (where we set the default `path` to the `loc` defined in section 1.2). For Part A of the function, we can conditionally set what image analysis we are interested in and set the `exnfile` object to the correct file extension. We have listed each image analysis and the argument name to be specified as the filetype in Table 2. As the last conditional (at the `else` level), we have used the `stop` function which will occur if `filetype` has not been specified, and will report an error. Part B is the same as in section 3.1 above however it is worth noting that we used `merge` here instead of `rbind()` (section 2.1) because this allows our function to handle data in both long and wide format. Then, in Part C, we use the `return` function to output our newly imported data.

Table 2: Table 2. QCPA image analysis and the different filetype arguments

Image analysis	Filetype argument
VCA,BSA,CAA	"VCA"
Cluster Particle Analysis	"PartAn"
Cluster Results	"Clust"
Individual Particle	"IndParticle"

```r
readQCPA <- function(Ani_ID_dat, filetype = "NA", path = loc) {
  # loc is assigned by default, but can be changed

  # Part A:
  # Here, we specify arguments for different image analyses so that we can
  # extract the correct file extension
  if (filetype == "VCA") {
    exnfile <- "_Summary Results.csv"
  } else if (filetype == "PartAn") {
    exnfile <- "_Cluster Particle Analysis Summary Results.csv"
  } else if (filetype == "Clust") {
    exnfile <- "_Cluster Results.csv"
  } else if (filetype == "IndParticle") {
    exnfile <- "_Individual Particle Results.csv"
  } else {
    stop('Specify analysis output type: e.g.,
         filetype =\n"VCA"\n"PartAn"\n"Clust"\n"IndParticle"')
  } # stop command halts the function and specifies how the user went wrong

  # Part B:
  # Once the file extension condition is met, we use lapply to extract the data,
  # using a nested function within lapply
  data_list <- lapply(1:nrow(Ani_ID_dat), function(x) { # for each row in Ani_ID_dat...
```

```r
    # extract the individual row information
    tmpLoc <- paste(unlist(Ani_ID_dat[x, ]), collapse = "/")

    # create a directory string
    tmpLoc <- paste0(paste(path, tmpLoc, sep = "/"), exnfile)

    # read in the data using the newly specified location
    data <- read.csv(tmpLoc)

    # add the specific animal information to the data
    data <- merge(Ani_ID_dat[x, ], data)

    data # returns data
  })

  # Part C:
  # We then collapse the data_list list into a data frame by calling rbind
  # in the function do.call
  data_df <- do.call(rbind, data_list)

  return(data_df) # outputs the data frame
}
```

To demonstrate how the stop warning works, we have implemented the `readQCPA()` function here without specifying the image analysis file type.

```r
readQCPA(Animal_Info_ROI) # We include only the animal id data set
```

```
## Error in readQCPA(Animal_Info_ROI): Specify analysis output type: e.g.,
##          filetype =
## "VCA"
## "PartAn"
## "Clust"
## "IndParticle"
```

We can then import our data using our function and do some data cleaning by removing unnecessary columns.

```r
# VCA,BSA,CAA analysis:
# Read in and assign analysis
VCA_analysis <- readQCPA(Animal_Info_ROI, filetype = "VCA")

# head(VCA_analysis)# Inspect data

# Subset data excluding the column Image and X
VCA_analysis <- VCA_analysis[, !colnames(VCA_analysis) %in% c("X", "Image")]

dim(VCA_analysis) # 18 rows, by 131 columns
```

```
## [1]  18 131
```

```r
VCA_analysis[1:2, ] # Observe the first two rows of data
```

```
##             Species                              Ind Dist    ROI CAA.Sc CAA.Jc
## 1 Aphelodoris varia 3_AphelodorisVaria_10_NF_D_RAW 30cm animal      1      1
## 2 Aphelodoris varia 3_AphelodorisVaria_10_NF_D_RAW 50cm animal      1      1
##   CAA.St CAA.Jt CAA.Hc CAA.Qc CAA.Ht CAA.Qt CAA.Scpl CAA.Qcpl CAA.C CAA.Sc.Hrz
## 1    Inf    NaN      0    NaN      0    NaN      Inf      NaN     0          1
## 2    Inf    NaN      0    NaN      0    NaN      Inf      NaN     0          1
##   CAA.Sc.Vrt CAA.Jc.Hrz CAA.Jc.Vrt CAA.St.Hrz CAA.St.Vrt CAA.Jt.Hrz CAA.Jt.Vrt
## 1          1          1          1        Inf        Inf        NaN        NaN
## 2          1          1          1        Inf        Inf        NaN        NaN
##   CAA.Hc.Hrz CAA.Hc.Vrt CAA.Qc.Hrz CAA.Qc.Vrt CAA.Ht.Hrz CAA.Ht.Vrt CAA.Qt.Hrz
## 1          0          0        NaN        NaN          0          0        NaN
## 2          0          0        NaN        NaN          0          0        NaN
##   CAA.Qt.Vrt CAA.Scpl.Hrz CAA.Scpl.Vrt CAA.Qcpl.Hrz CAA.Qcpl.Vrt CAA.C.Hrz
## 1        NaN          Inf          Inf          NaN          NaN         0
## 2        NaN          Inf          Inf          NaN          NaN         0
##   CAA.C.Vrt CAA.PT CAA.PT.Hrz CAA.PT.Vrt CAA.Asp     VCA.ML VCA.sL VCA.CVL
## 1         0    Inf        Inf        Inf     NaN 0.07415875    NaN     NaN
## 2         0    Inf        Inf        Inf     NaN 0.07259081    NaN     NaN
##    VCA.MDmax VCA.sDmax VCA.CVDmax VCA.MSsat VCA.sSsat VCA.CVSsat VCA.MSL
## 1 0.08914595       NaN        NaN  2.695919       NaN        NaN     NaN
## 2 0.08963545       NaN        NaN  2.723553       NaN        NaN     NaN
##   VCA.sSL VCA.CVSL VCA.MS VCA.sS VCA.CVS VCA.ML.Hrz VCA.sL.Hrz VCA.CVL.Hrz
## 1     NaN      NaN    NaN    NaN     NaN 0.07415875        NaN         NaN
## 2     NaN      NaN    NaN    NaN     NaN 0.07259081        NaN         NaN
##   VCA.ML.Vrt VCA.sL.Vrt VCA.CVL.Vrt VCA.MDmax.Hrz VCA.sDmax.Hrz VCA.CVDmax.Hrz
## 1 0.07415875        NaN         NaN    0.08914595           NaN            NaN
## 2 0.07259081        NaN         NaN    0.08963545           NaN            NaN
##   VCA.MDmax.Vrt VCA.sDmax.Vrt VCA.CVDmax.Vrt VCA.MSsat.Hrz VCA.sSsat.Hrz
## 1    0.08914595           NaN            NaN      2.695919           NaN
## 2    0.08963545           NaN            NaN      2.723553           NaN
##   VCA.CVSsat.Hrz VCA.MSsat.Vrt VCA.sSsat.Vrt VCA.CVSsat.Vrt VCA.MSL.Hrz
## 1            NaN      2.695919           NaN            NaN         NaN
## 2            NaN      2.723553           NaN            NaN         NaN
##   VCA.sSL.Hrz VCA.CVSL.Hrz VCA.MSL.Vrt VCA.sSL.Vrt VCA.CVSL.Vrt VCA.MS.Hrz
## 1         NaN          NaN         NaN         NaN          NaN        NaN
## 2         NaN          NaN         NaN         NaN          NaN        NaN
##   VCA.sS.Hrz VCA.CVS.Hrz VCA.MS.Vrt VCA.sS.Vrt VCA.CVS.Vrt BSA.BML BSA.BsL
## 1        NaN         NaN        NaN        NaN         NaN       0       0
## 2        NaN         NaN        NaN        NaN         NaN       0       0
##   BSA.BCVL BSA.BMDmax BSA.BsDmax BSA.BCVDmax BSA.BMSsat BSA.BsSsat BSA.BCVSsat
## 1      NaN          0          0         NaN          0          0         NaN
## 2      NaN          0          0         NaN          0          0         NaN
##   BSA.BMSL BSA.BsSL BSA.BCVSL BSA.BMS BSA.BsS BSA.BCVS BSA.BML.Hrz BSA.BsL.Hrz
## 1        0        0       NaN       0       0      NaN           0           0
## 2        0        0       NaN       0       0      NaN           0           0
##   BSA.BCVL.Hrz BSA.BML.Vrt BSA.BsL.Vrt BSA.BCVL.Vrt BSA.BMDmax.Hrz
## 1          NaN           0           0          NaN              0
## 2          NaN           0           0          NaN              0
##   BSA.BsDmax.Hrz BSA.BCVDmax.Hrz BSA.BMDmax.Vrt BSA.BsDmax.Vrt BSA.BCVDmax.Vrt
## 1              0             NaN              0              0             NaN
## 2              0             NaN              0              0             NaN
```

```
##   BSA.BMSsat.Hrz BSA.BsSsat.Hrz BSA.BCVSsat.Hrz BSA.BMSsat.Vrt BSA.BsSsat.Vrt
## 1              0              0             NaN              0              0
## 2              0              0             NaN              0              0
##   BSA.BCVSsat.Vrt BSA.BMSL.Hrz BSA.BsSL.Hrz BSA.BCVSL.Hrz BSA.BMSL.Vrt
## 1             NaN            0            0           NaN            0
## 2             NaN            0            0           NaN            0
##   BSA.BsSL.Vrt BSA.BCVSL.Vrt BSA.BMS.Hrz BSA.BsS.Hrz BSA.BCVS.Hrz BSA.BMS.Vrt
## 1            0           NaN           0           0          NaN           0
## 2            0           NaN           0           0          NaN           0
##   BSA.BsS.Vrt BSA.BCVS.Vrt
## 1           0          NaN
## 2           0          NaN
```

```r
# Cluster Particle Analysis:
# Read in and assign analysis
Particle_analysis <- readQCPA(Animal_Info_ROI, filetype = "PartAn")

# head(Particle_analysis)# Inspect data

dim(Particle_analysis) # 52 rows, by 20 columns
```

```
## [1] 52 20
```

```r
colnames(Particle_analysis)[5]
```

```
## [1] "Cluster.ID"
```

```r
# Change column name to be consistent with Cluster analysis
colnames(Particle_analysis)[5] <- "ClusterID"

Particle_analysis[1:2, ] # Observe the first two rows of data
```

```
##             Species                              Ind Dist    ROI ClusterID Count
## 1 Aphelodoris varia 3_AphelodorisVaria_10_NF_D_RAW 30cm animal         1     4
## 2 Aphelodoris varia 3_AphelodorisVaria_10_NF_D_RAW 50cm animal         1     7
##   Total.Area Average.Size   X.Area    Perim.    Major     Minor    Angle
## 1       2953     738.2500 24.46764 204.91169 52.96716 17.837978 79.13099
## 2       1084     154.8571 24.31038  66.67049 19.45619  6.734171 60.18813
##       Circ.  Solidity    Feret   FeretX   FeretY FeretAngle MinFeret
## 1 0.2135263 0.5150810 83.09385 19.25000 60.25000   79.74821 31.79423
## 2 0.4806954 0.7693886 27.63505 23.42857 30.42857  105.63732 10.86412
```

```r
# Cluster Results:
# Read in and assign analysis
Cluster_analysis <- readQCPA(Animal_Info_ROI, filetype = "Clust")

# head(Cluster_Analysis)# inspect data

# Subset data excluding the column Image and X
Cluster_analysis <- Cluster_analysis[, !colnames(Cluster_analysis) %in% c("X", "Image")]

dim(Cluster_analysis) # 52 rows, by 21 columns
```

```
## [1] 52 21
```

```r
Cluster_analysis[1:2, ] # Observe the first two rows of data
```

```
##             Species                            Ind Dist    ROI Pass ClusterID
## 1 Aphelodoris varia 3_AphelodorisVaria_10_NF_D_RAW 30cm animal    7         1
## 2 Aphelodoris varia 3_AphelodorisVaria_10_NF_D_RAW 50cm animal    6         1
##      lw_mean      lw_sd    mw_mean      mw_sd    sw_mean      sw_sd   dbl_mean
## 1 0.07855250 0.01874494 0.06783328 0.01802219 0.06569355 0.01396244 0.07415875
## 2 0.07697873 0.01535109 0.06627374 0.01422326 0.06431390 0.01060753 0.07259081
##       dbl_sd Area Coverage..pc.    RNL.X     RNL.Y RNL.Saturation
## 1 0.01840701 9116           100 2.074853 -1.721327       2.695919
## 2 0.01485467 3375           100 2.117579 -1.712775       2.723553
##   Dmax.Chromaticity Dmax.channel
## 1        0.08914595        lw:sw
## 2        0.08963545        lw:sw
```

```r
# Individual Particle Analysis:
# Read in and assign analysis
IndParticle_analysis <- readQCPA(Animal_Info_ROI, filetype = "IndParticle")

# head(Cluster_Analysis)# inspect data

# Subset data excluding the column X.1
IndParticle_analysis <- IndParticle_analysis[, colnames(IndParticle_analysis) != "X.1"]

dim(IndParticle_analysis) # 143 rows, by 23 columns
```

```
## [1] 143  23
```

```r
IndParticle_analysis[1:2, ] # Observe the first two rows of data
```

```
##             Species                            Ind Dist    ROI
## 1 Aphelodoris varia 3_AphelodorisVaria_10_NF_D_RAW 30cm animal
## 2 Aphelodoris varia 3_AphelodorisVaria_10_NF_D_RAW 30cm animal
##   ClusterID_Particle.         X        Y        XM       YM   Perim.    Major
## 1                 1_1  9.001362 10.64169  9.001362 10.64169 160.6690 29.94767
## 2                 1_2 71.712538 16.22783 71.712538 16.22783 173.4975 45.31033
##      Minor     Angle     Circ.    Feret FeretX FeretY FeretAngle MinFeret
## 1 15.60318  47.74661 0.1786535 60.82763      0     42   43.66778 31.08705
## 2 18.37768 109.97441 0.2730243 71.11259     50      0  115.84439 28.42651
##         AR     Round  Solidity Area..px.
## 1 1.919332 0.5210147 0.3797206       367
## 2 2.465509 0.4055958 0.6285440       654
```

4 Get a .csv list from LEIA output

Here, we implement a function similar to section 3.1. However, because QCPA batch script exports LEIA analysis into a subdirectory, we have to add two additional folders to our location, `LEIA` and the ROI as a folder. For the LEIA folder, as we iterate over each row from `Animal_Info_ROI`, we convert the row into a character vector and add 'LEIA' as an element. We include a backslash to treat the ROI as a folder before specifying the image analysis file extension, `_Local Edge Intensity Analysis.csv` (Table 1).

```r
LEIA_Res_list <- lapply(
  1:nrow(Animal_Info_ROI), # For each row in Animal_Info_ROI
  function(x) { # Start of our nested function

    # Convert row into a character vector
    tmpLoc <- as.character(unlist(Animal_Info_ROI[x, ]))

    # Add and order the new sub folder LEIA before ROI
    tmpLoc <- c(tmpLoc, "LEIA")[c(1:3, 5, 4)]

    # Paste tmploc into a directory character
    tmpLoc <- paste(tmpLoc, collapse = "/")

    # Create a directory string
    tmpLoc <- paste0(
      paste(loc, tmpLoc, sep = "/"),
      "/_Local Edge Intensity Analysis.csv"
    )

    # Read in the data using the newly specified location
    data <- read.csv(tmpLoc)

    # Add the specific animal information to the data
    data <- merge(Animal_Info_ROI[x, ], data)

    data # returns data
  }
)

# We then collapse the LEIA_Res_list list into a data frame by
# calling rbind in the function do.call
LEIA_Res_analysis <- do.call(rbind, LEIA_Res_list)

# Subset data excluding the column Image and X
LEIA_Res_analysis <- LEIA_Res_analysis[, !colnames(LEIA_Res_analysis) %in% c("X", "Image")]

# Check the imported data
dim(LEIA_Res_analysis) # 18 rows by 37 columns
```

```
## [1] 18 37
```

```r
LEIA_Res_analysis[1:2, ] # Observe the first two rows of data
```

```
##             Species                            Ind Dist    ROI  Col.mean
## 1 Aphelodoris varia 3_AphelodorisVaria_10_NF_D_RAW 30cm animal 0.1736449
## 2 Aphelodoris varia 3_AphelodorisVaria_10_NF_D_RAW 50cm animal 0.1663600
##      Col.sd  Col.CoV Col.skew Col.kurtosis  Lum.mean   Lum.sd  Lum.CoV Lum.skew
## 1 0.2851837 1.642339 2.351301     9.700354 0.8928419 1.431054 1.602808 2.255974
## 2 0.2695337 1.620183 2.286320     9.671851 0.8987935 1.380676 1.536144 2.060313
##   Lum.kurtosis Col.mean.hrz Col.sd.hrz Col.CoV.hrz Col.skew.hrz
## 1     9.558368   0.08770301  0.1856711    2.117043     3.926095
## 2     8.282279   0.08205933  0.1756506    2.140532     3.887552
```

```
##   Col.kurtosis.hrz Col.mean.vrt Col.sd.vrt Col.CoV.vrt Col.skew.vrt
## 1         22.61682   0.08907545  0.1942661    2.180916     3.913759
## 2         22.67175   0.08694090  0.1873940    2.155418     3.881844
##   Col.kurtosis.vrt Lum.mean.hrz Lum.sd.hrz Lum.CoV.hrz Lum.skew.hrz
## 1         22.07321    0.4355562  0.8958863    2.056879     3.612730
## 2         22.90526    0.4406635  0.8761384    1.988225     3.493638
##   Lum.kurtosis.hrz Lum.mean.vrt Lum.sd.vrt Lum.CoV.vrt Lum.skew.vrt
## 1         20.01520    0.4563516  0.9904756    2.170422     3.852029
## 2         19.45551    0.4573795  0.9806892    2.144148     3.690217
##   Lum.kurtosis.vrt Transform Col.thresh Lum.thresh
## 1         22.42973      none          0          0
## 2         20.01364      none          0          0
```

5 Get a .csv list from GabRat output

For GabRat analyses, there are no specific ROIs, so we only need the first three columns from `Animal_Info_ROI`. Similar to LEIA (section 4, above), QCPA batch script also exports GabRat analysis into a subdirectory. We will again iterate over each row of `Animal_Info_ROI` and convert the row into a character vector, adding 'GabRat' as an element. From Table 1, the GabRat file extension is: `_GabRat_Results.csv`, where we place a backslash before the file name to specify that it is located within the GabRat folder.

For this work example, we are only interested in the GabRat values `dbl` (the luminance channel of the visual system considered in the analysis), so we will then extract the rows with only these values. We then extract the ROI information in a method similar to section 2.3.

```r
# Because we are not using ROI which adds three levels (rows),
# we trim data frame using unique()
# For each the 6 rows in Animal_Info_ROI
GabRat_Res_list <- lapply(
  1:nrow(unique(Animal_Info_ROI[, 1:3])),
  function(x) { # Start of our nested function

    # Using the first three columns, convert row into a character vector
    tmpLoc <- as.character(unlist(Animal_Info_ROI[x, 1:3]))

    tmpLoc <- c(tmpLoc, "GabRat") # Add the new sub folder GabRat

    # Paste tmploc into a directory character
    tmpLoc <- paste(tmpLoc, collapse = "/")

    # Create a directory string
    tmpLoc <- paste0(paste(loc, tmpLoc, sep = "/"), "/_GabRat_Results.csv")

    # Read in the data using the newly specified location
    data <- read.csv(tmpLoc)

    # Add the specific animal information to the data,
    # using only the first three columns
    data <- merge(Animal_Info_ROI[x, 1:3], data)

    data # Returns data
  }
```

```r
)

# We then collapse the LEIA_Res_list list into a data frame by calling
# rbind in the function do.call
GabRat_Res_analysis <- do.call(rbind, GabRat_Res_list)

# Check the imported data
dim(GabRat_Res_analysis) # 72 rows by 7 columns
```

```
## [1] 72  7
```

```r
head(GabRat_Res_analysis) # We only want every ~ fourth row for the dbl values
```

```
##             Species                            Ind Dist X
## 1 Aphelodoris varia 3_AphelodorisVaria_10_NF_D_RAW 30cm 1
## 2 Aphelodoris varia 3_AphelodorisVaria_10_NF_D_RAW 30cm 2
## 3 Aphelodoris varia 3_AphelodorisVaria_10_NF_D_RAW 30cm 3
## 4 Aphelodoris varia 3_AphelodorisVaria_10_NF_D_RAW 30cm 4
## 5 Aphelodoris varia 3_AphelodorisVaria_10_NF_D_RAW 30cm 5
## 6 Aphelodoris varia 3_AphelodorisVaria_10_NF_D_RAW 30cm 6
##                                                                                ID
## 1            3_AphelodorisVaria_10_NF_D_RAW.ORF_Acuity1_Dist300_WholeImage_animal_lw
## 2            3_AphelodorisVaria_10_NF_D_RAW.ORF_Acuity1_Dist300_WholeImage_animal_mw
## 3            3_AphelodorisVaria_10_NF_D_RAW.ORF_Acuity1_Dist300_WholeImage_animal_sw
## 4           3_AphelodorisVaria_10_NF_D_RAW.ORF_Acuity1_Dist300_WholeImage_animal_dbl
## 5 3_AphelodorisVaria_10_NF_D_RAW.ORF_Acuity1_Dist300_WholeImage_animal+background_lw
## 6 3_AphelodorisVaria_10_NF_D_RAW.ORF_Acuity1_Dist300_WholeImage_animal+background_mw
##   Sigma    GabRat
## 1     3 0.1666780
## 2     3 0.1774265
## 3     3 0.1902168
## 4     3 0.1682120
## 5     3 0.5320508
## 6     3 0.5181680
```

```r
# Set up a true/false condition on whether row has the pattern "_dbl",
# then subset data on this condition
GabRat_Res_analysis <- GabRat_Res_analysis[endsWith(GabRat_Res_analysis$ID, "_dbl"), ]
rownames(GabRat_Res_analysis) <- NULL # Reset row numbers

# Then we want to determine ROI where we split the ID column
# data on the shared pattern "WholeImage_"
# We then only take the second elemnt, using a FALSE,TRUE index
GabRat_Res_analysis$ROI <- unlist(
  strsplit(GabRat_Res_analysis$ID, "WholeImage_")
)[c(FALSE, TRUE)]

# Then remove the '_dbl'
GabRat_Res_analysis$ROI <- gsub("_dbl", "", GabRat_Res_analysis$ROI)

# Remove unnecessary columns, subset data excluding the column X
GabRat_Res_analysis <- GabRat_Res_analysis[, !colnames(GabRat_Res_analysis)
```

```r
%in% c("X", "ID", "Sigma")]

# Check the imported data
dim(GabRat_Res_analysis) # 18 rows by 5 columns
```

```
## [1] 18  5
```

```r
GabRat_Res_analysis[1:2, ] # Observe the first two rows of data
```

```
##             Species                            Ind Dist    GabRat
## 1 Aphelodoris varia 3_AphelodorisVaria_10_NF_D_RAW 30cm 0.1682120
## 2 Aphelodoris varia 3_AphelodorisVaria_10_NF_D_RAW 30cm 0.5332068
##                 ROI
## 1            animal
## 2 animal+background
```

6 Combine data sets and data preparation

6.1 Merging data sets

To merge files, we will again use the `merge()` function.

6.1.1 Particle analyses

```r
# Merging two files together
Cluster_Particle_analysis <- merge(Cluster_analysis, Particle_analysis,
  by = c("Species", "Ind", "Dist", "ROI", "ClusterID")
)

dim(Cluster_Particle_analysis) # 52 rows and 36 columns
```

```
## [1] 52 36
```

```r
Cluster_Particle_analysis[1:2, ] # Inspect data
```

```
##             Species                            Ind Dist               ROI
## 1 Aphelodoris varia 3_AphelodorisVaria_10_NF_D_RAW 30cm            animal
## 2 Aphelodoris varia 3_AphelodorisVaria_10_NF_D_RAW 30cm animal+background
##   ClusterID Pass    lw_mean       lw_sd    mw_mean       mw_sd    sw_mean
## 1         1    7 0.07855250 0.018744942 0.06783328 0.018022186 0.06569355
## 2         1   20 0.03184927 0.006353116 0.02877177 0.004386119 0.03404231
##         sw_sd   dbl_mean      dbl_sd  Area Coverage..pc.    RNL.X     RNL.Y
## 1 0.013962437 0.07415875 0.018407013  9116     100.00000 2.074853 -1.721327
## 2 0.006827025 0.03058787 0.005537348 32133      71.76229 1.437122  1.917127
##   RNL.Saturation Dmax.Chromaticity Dmax.channel Count Total.Area Average.Size
## 1       2.695919        0.08914595        lw:sw     4       2953       738.25
## 2       2.395975        0.08390701        mw:sw     2      32133     16066.50
```

```
##     X.Area   Perim.     Major     Minor    Angle     Circ.  Solidity     Feret
## 1 24.46764 204.9117  52.96716  17.83798 79.13099 0.2135263 0.5150810  83.09385
## 2 45.13759 740.6488 113.31427 101.73291 73.06783 0.4666892 0.7740756 150.38945
##   FeretX FeretY FeretAngle  MinFeret
## 1  19.25  60.25   79.74821  31.79423
## 2  51.50 173.00   91.85460 119.27169
```

6.1.2 VCA,BSA,CAA data, LEIA output and GabRat analyses

Here, we demonstrate how to collate data across the VCA, BSA, CAA data, LEIA output, and GabRat output. We have excluded the particle analysis outcome from this step.

```r
# Merge the first two data frames (VCA_analysis and LEIA_Res_analysis) together
VCA_LEIA_GabRat_analysis <- merge(VCA_analysis, LEIA_Res_analysis,
  by = c("Species", "Ind", "Dist", "ROI")
)

VCA_LEIA_GabRat_analysis[1:2, ] # Inspect data
```

```
##             Species                             Ind Dist               ROI
## 1 Aphelodoris varia 3_AphelodorisVaria_10_NF_D_RAW 30cm            animal
## 2 Aphelodoris varia 3_AphelodorisVaria_10_NF_D_RAW 30cm animal+background
##     CAA.Sc    CAA.Jc   CAA.St    CAA.Jt  CAA.Hc    CAA.Qc    CAA.Ht    CAA.Qt
## 1 1.000000 1.0000000      Inf       NaN 0.00000       NaN 0.0000000       NaN
## 2 1.763881 0.5879602 2.098908 0.6996359 0.76058 0.6923097 0.8373184 0.7621601
##    CAA.Scpl  CAA.Qcpl      CAA.C CAA.Sc.Hrz CAA.Sc.Vrt CAA.Jc.Hrz CAA.Jc.Vrt
## 1       Inf       NaN 0.00000000   1.000000   1.000000  1.0000000  1.0000000
## 2 0.6437981 0.7272349 0.01618755   1.763113   1.764646  0.5877043  0.5882152
##   CAA.St.Hrz CAA.St.Vrt CAA.Jt.Hrz CAA.Jt.Vrt CAA.Hc.Hrz CAA.Hc.Vrt CAA.Qc.Hrz
## 1        Inf        Inf        NaN        NaN    0.00000  0.0000000        NaN
## 2   2.069081   2.118686  0.6896936  0.7062285    0.76029  0.7608685  0.6920457
##   CAA.Qc.Vrt CAA.Ht.Hrz CAA.Ht.Vrt CAA.Qt.Hrz CAA.Qt.Vrt CAA.Scpl.Hrz
## 1        NaN  0.0000000  0.0000000        NaN        NaN          Inf
## 2  0.6925724  0.8386991  0.8298446  0.7634168  0.7553571    0.6386989
##   CAA.Scpl.Vrt CAA.Qcpl.Hrz CAA.Qcpl.Vrt  CAA.C.Hrz  CAA.C.Vrt   CAA.PT
## 1          Inf          NaN          NaN 0.00000000 0.00000000      Inf
## 2    0.6472219    0.7277313    0.7239647 0.01768539 0.01469036 60.77585
##   CAA.PT.Hrz CAA.PT.Vrt   CAA.Asp     VCA.ML    VCA.sL   VCA.CVL  VCA.MDmax
## 1        Inf        Inf       NaN 0.07415875       NaN       NaN 0.08914595
## 2   55.54384   67.07187 0.4529912 0.04150538 0.0231051 0.5566772 0.08559733
##     VCA.sDmax VCA.CVDmax VCA.MSsat VCA.sSsat VCA.CVSsat  VCA.MSL VCA.sSL
## 1         NaN        NaN  2.695919       NaN        NaN      NaN     NaN
## 2 0.009909556  0.1157695  2.469105 0.2776953   0.112468 16.01608 6.01111
##    VCA.CVSL   VCA.MS   VCA.sS   VCA.CVS VCA.ML.Hrz VCA.sL.Hrz VCA.CVL.Hrz
## 1       NaN      NaN      NaN       NaN 0.07415875        NaN         NaN
## 2 0.3753171 3.334727 1.209654 0.3627445 0.04129542 0.02296875   0.5562057
##   VCA.ML.Vrt VCA.sL.Vrt VCA.CVL.Vrt VCA.MDmax.Hrz VCA.sDmax.Hrz VCA.CVDmax.Hrz
## 1 0.07415875        NaN         NaN    0.08914595           NaN            NaN
## 2 0.04130759  0.0229682   0.5560287    0.08557285   0.009828942      0.1148605
##   VCA.MDmax.Vrt VCA.sDmax.Vrt VCA.CVDmax.Vrt VCA.MSsat.Hrz VCA.sSsat.Hrz
## 1    0.08914595           NaN            NaN      2.695919           NaN
## 2    0.08556912   0.009826575      0.1148379      2.467878     0.2756307
##   VCA.CVSsat.Hrz VCA.MSsat.Vrt VCA.sSsat.Vrt VCA.CVSsat.Vrt VCA.MSL.Hrz
```

```
## 1            NaN      2.695919           NaN            NaN         NaN
## 2      0.1116873      2.467834     0.2755308      0.1116488    16.01783
##   VCA.sSL.Hrz VCA.CVSL.Hrz VCA.MSL.Vrt VCA.sSL.Vrt VCA.CVSL.Vrt VCA.MS.Hrz
## 1         NaN          NaN         NaN         NaN          NaN        NaN
## 2    6.010669    0.3752486    16.01434    6.011548    0.3753854    3.33508
##   VCA.sS.Hrz VCA.CVS.Hrz VCA.MS.Vrt VCA.sS.Vrt VCA.CVS.Vrt   BSA.BML    BSA.BsL
## 1        NaN         NaN        NaN        NaN         NaN 0.0000000 0.00000000
## 2   1.209564   0.3626792   3.334376   1.209744   0.3628096 0.2888441 0.09471583
##    BSA.BCVL BSA.BMDmax BSA.BsDmax BSA.BCVDmax BSA.BMSsat BSA.BsSsat BSA.BCVSsat
## 1       NaN 0.00000000 0.00000000         NaN 0.00000000 0.00000000         NaN
## 2 0.3279133 0.07865894 0.08357918    1.062552 0.06068302 0.08350352    1.376061
##   BSA.BMSL BSA.BsSL BSA.BCVSL  BSA.BMS   BSA.BsS BSA.BCVS BSA.BML.Hrz
## 1  0.00000 0.000000       NaN 0.000000 0.0000000      NaN   0.0000000
## 2 11.98163 4.230545 0.3530861 2.522453 0.8547204 0.338845   0.2939087
##   BSA.BsL.Hrz BSA.BCVL.Hrz BSA.BML.Vrt BSA.BsL.Vrt BSA.BCVL.Vrt BSA.BMDmax.Hrz
## 1  0.00000000          NaN   0.0000000  0.00000000          NaN     0.00000000
## 2  0.09529334    0.3242277   0.2827497  0.09347208    0.3305825     0.07536487
##   BSA.BsDmax.Hrz BSA.BCVDmax.Hrz BSA.BMDmax.Vrt BSA.BsDmax.Vrt BSA.BCVDmax.Vrt
## 1     0.00000000             NaN     0.00000000     0.00000000             NaN
## 2     0.08251804        1.094914     0.08262291     0.08458379        1.023733
##   BSA.BMSsat.Hrz BSA.BsSsat.Hrz BSA.BCVSsat.Hrz BSA.BMSsat.Vrt BSA.BsSsat.Vrt
## 1     0.00000000      0.0000000             NaN     0.00000000     0.00000000
## 2     0.05763893      0.0828074        1.436658     0.06434617     0.08411479
##   BSA.BCVSsat.Vrt BSA.BMSL.Hrz BSA.BsSL.Hrz BSA.BCVSL.Hrz BSA.BMSL.Vrt
## 1             NaN      0.00000      0.00000           NaN      0.00000
## 2        1.307223     12.20617      4.27166     0.3499591     11.71142
##   BSA.BsSL.Vrt BSA.BCVSL.Vrt BSA.BMS.Hrz BSA.BsS.Hrz BSA.BCVS.Hrz BSA.BMS.Vrt
## 1     0.000000           NaN    0.000000   0.0000000          NaN    0.000000
## 2     4.156481     0.3549083    2.567851   0.8627548    0.3359832    2.467822
##   BSA.BsS.Vrt BSA.BCVS.Vrt  Col.mean    Col.sd  Col.CoV Col.skew Col.kurtosis
## 1   0.0000000          NaN 0.1736449 0.2851837 1.642339 2.351301     9.700354
## 2   0.8400862    0.3404161 0.2161124 0.4168882 1.929034 3.158602    19.114901
##    Lum.mean   Lum.sd  Lum.CoV Lum.skew Lum.kurtosis Col.mean.hrz Col.sd.hrz
## 1 0.8928419 1.431054 1.602808 2.255974     9.558368   0.08770301  0.1856711
## 2 0.7599581 1.631294 2.146557 4.713521    38.984497   0.11245671  0.2857060
##   Col.CoV.hrz Col.skew.hrz Col.kurtosis.hrz Col.mean.vrt Col.sd.vrt Col.CoV.vrt
## 1    2.117043     3.926095         22.61682   0.08907545  0.1942661    2.180916
## 2    2.540587     5.473994         47.54446   0.10647940  0.2643248    2.482403
##   Col.skew.vrt Col.kurtosis.vrt Lum.mean.hrz Lum.sd.hrz Lum.CoV.hrz
## 1     3.913759         22.07321    0.4355562  0.8958863    2.056879
## 2     5.224416         45.33121    0.3860825  1.0964629    2.839970
##   Lum.skew.hrz Lum.kurtosis.hrz Lum.mean.vrt Lum.sd.vrt Lum.CoV.vrt
## 1     3.612730         20.01520    0.4563516  0.9904756    2.170422
## 2     7.414604         84.33807    0.3751880  1.0664402    2.842415
##   Lum.skew.vrt Lum.kurtosis.vrt Transform Col.thresh Lum.thresh
## 1     3.852029         22.42973      none          0          0
## 2     7.844373         98.05338      none          0          0
```

```r
dim(VCA_LEIA_GabRat_analysis) # 18 rows and 164 columns
```

```
## [1]  18 164
```

```r
# Merge the new data frame (VCA_LEIA_GabRat_analysis) with GabRat_Res_analysis
VCA_LEIA_GabRat_analysis <- merge(VCA_LEIA_GabRat_analysis, GabRat_Res_analysis,
  by = c("Species", "Ind", "Dist", "ROI")
)

VCA_LEIA_GabRat_analysis[1:2, ] # Inspect data again
```

```
##             Species                              Ind Dist               ROI
## 1 Aphelodoris varia 3_AphelodorisVaria_10_NF_D_RAW 30cm            animal
## 2 Aphelodoris varia 3_AphelodorisVaria_10_NF_D_RAW 30cm animal+background
##     CAA.Sc    CAA.Jc   CAA.St    CAA.Jt  CAA.Hc    CAA.Qc    CAA.Ht    CAA.Qt
## 1 1.000000 1.0000000      Inf       NaN 0.00000       NaN 0.0000000       NaN
## 2 1.763881 0.5879602 2.098908 0.6996359 0.76058 0.6923097 0.8373184 0.7621601
##    CAA.Scpl  CAA.Qcpl      CAA.C CAA.Sc.Hrz CAA.Sc.Vrt CAA.Jc.Hrz CAA.Jc.Vrt
## 1       Inf       NaN 0.00000000   1.000000   1.000000  1.0000000  1.0000000
## 2 0.6437981 0.7272349 0.01618755   1.763113   1.764646  0.5877043  0.5882152
##   CAA.St.Hrz CAA.St.Vrt CAA.Jt.Hrz CAA.Jt.Vrt CAA.Hc.Hrz CAA.Hc.Vrt CAA.Qc.Hrz
## 1        Inf        Inf        NaN        NaN    0.00000  0.0000000        NaN
## 2   2.069081   2.118686  0.6896936  0.7062285    0.76029  0.7608685  0.6920457
##   CAA.Qc.Vrt CAA.Ht.Hrz CAA.Ht.Vrt CAA.Qt.Hrz CAA.Qt.Vrt CAA.Scpl.Hrz
## 1        NaN  0.0000000  0.0000000        NaN        NaN          Inf
## 2  0.6925724  0.8386991  0.8298446  0.7634168  0.7553571    0.6386989
##   CAA.Scpl.Vrt CAA.Qcpl.Hrz CAA.Qcpl.Vrt  CAA.C.Hrz  CAA.C.Vrt   CAA.PT
## 1          Inf          NaN          NaN 0.00000000 0.00000000      Inf
## 2    0.6472219    0.7277313    0.7239647 0.01768539 0.01469036 60.77585
##   CAA.PT.Hrz CAA.PT.Vrt   CAA.Asp     VCA.ML    VCA.sL   VCA.CVL  VCA.MDmax
## 1        Inf        Inf       NaN 0.07415875       NaN       NaN 0.08914595
## 2   55.54384   67.07187 0.4529912 0.04150538 0.0231051 0.5566772 0.08559733
##     VCA.sDmax VCA.CVDmax VCA.MSsat VCA.sSsat VCA.CVSsat  VCA.MSL VCA.sSL
## 1         NaN        NaN  2.695919       NaN        NaN      NaN     NaN
## 2 0.009909556  0.1157695  2.469105 0.2776953   0.112468 16.01608 6.01111
##    VCA.CVSL   VCA.MS   VCA.sS   VCA.CVS VCA.ML.Hrz VCA.sL.Hrz VCA.CVL.Hrz
## 1       NaN      NaN      NaN       NaN 0.07415875        NaN         NaN
## 2 0.3753171 3.334727 1.209654 0.3627445 0.04129542 0.02296875   0.5562057
##   VCA.ML.Vrt VCA.sL.Vrt VCA.CVL.Vrt VCA.MDmax.Hrz VCA.sDmax.Hrz VCA.CVDmax.Hrz
## 1 0.07415875        NaN         NaN    0.08914595           NaN            NaN
## 2 0.04130759  0.0229682   0.5560287    0.08557285   0.009828942      0.1148605
##   VCA.MDmax.Vrt VCA.sDmax.Vrt VCA.CVDmax.Vrt VCA.MSsat.Hrz VCA.sSsat.Hrz
## 1    0.08914595           NaN            NaN      2.695919           NaN
## 2    0.08556912   0.009826575      0.1148379      2.467878     0.2756307
##   VCA.CVSsat.Hrz VCA.MSsat.Vrt VCA.sSsat.Vrt VCA.CVSsat.Vrt VCA.MSL.Hrz
## 1            NaN      2.695919           NaN            NaN         NaN
## 2      0.1116873      2.467834     0.2755308      0.1116488    16.01783
##   VCA.sSL.Hrz VCA.CVSL.Hrz VCA.MSL.Vrt VCA.sSL.Vrt VCA.CVSL.Vrt VCA.MS.Hrz
## 1         NaN          NaN         NaN         NaN          NaN        NaN
## 2    6.010669    0.3752486    16.01434    6.011548    0.3753854    3.33508
##   VCA.sS.Hrz VCA.CVS.Hrz VCA.MS.Vrt VCA.sS.Vrt VCA.CVS.Vrt   BSA.BML    BSA.BsL
## 1        NaN         NaN        NaN        NaN         NaN 0.0000000 0.00000000
## 2   1.209564   0.3626792   3.334376   1.209744   0.3628096 0.2888441 0.09471583
##    BSA.BCVL BSA.BMDmax BSA.BsDmax BSA.BCVDmax BSA.BMSsat BSA.BsSsat BSA.BCVSsat
## 1       NaN 0.00000000 0.00000000         NaN 0.00000000 0.00000000         NaN
## 2 0.3279133 0.07865894 0.08357918    1.062552 0.06068302 0.08350352    1.376061
##   BSA.BMSL BSA.BsSL BSA.BCVSL  BSA.BMS   BSA.BsS BSA.BCVS BSA.BML.Hrz
```

```
## 1  0.00000 0.000000       NaN 0.000000 0.0000000      NaN   0.0000000
## 2 11.98163 4.230545 0.3530861 2.522453 0.8547204 0.338845   0.2939087
##   BSA.BsL.Hrz BSA.BCVL.Hrz BSA.BML.Vrt BSA.BsL.Vrt BSA.BCVL.Vrt BSA.BMDmax.Hrz
## 1  0.00000000          NaN   0.0000000  0.00000000          NaN     0.00000000
## 2  0.09529334    0.3242277   0.2827497  0.09347208    0.3305825     0.07536487
##   BSA.BsDmax.Hrz BSA.BCVDmax.Hrz BSA.BMDmax.Vrt BSA.BsDmax.Vrt BSA.BCVDmax.Vrt
## 1     0.00000000             NaN     0.00000000     0.00000000             NaN
## 2     0.08251804        1.094914     0.08262291     0.08458379        1.023733
##   BSA.BMSsat.Hrz BSA.BsSsat.Hrz BSA.BCVSsat.Hrz BSA.BMSsat.Vrt BSA.BsSsat.Vrt
## 1     0.00000000      0.0000000             NaN     0.00000000     0.00000000
## 2     0.05763893      0.0828074        1.436658     0.06434617     0.08411479
##   BSA.BCVSsat.Vrt BSA.BMSL.Hrz BSA.BsSL.Hrz BSA.BCVSL.Hrz BSA.BMSL.Vrt
## 1             NaN      0.00000      0.00000           NaN      0.00000
## 2        1.307223     12.20617      4.27166     0.3499591     11.71142
##   BSA.BsSL.Vrt BSA.BCVSL.Vrt BSA.BMS.Hrz BSA.BsS.Hrz BSA.BCVS.Hrz BSA.BMS.Vrt
## 1     0.000000           NaN    0.000000   0.0000000          NaN    0.000000
## 2     4.156481     0.3549083    2.567851   0.8627548    0.3359832    2.467822
##   BSA.BsS.Vrt BSA.BCVS.Vrt  Col.mean    Col.sd  Col.CoV Col.skew Col.kurtosis
## 1   0.0000000          NaN 0.1736449 0.2851837 1.642339 2.351301     9.700354
## 2   0.8400862    0.3404161 0.2161124 0.4168882 1.929034 3.158602    19.114901
##    Lum.mean   Lum.sd  Lum.CoV Lum.skew Lum.kurtosis Col.mean.hrz Col.sd.hrz
## 1 0.8928419 1.431054 1.602808 2.255974     9.558368   0.08770301  0.1856711
## 2 0.7599581 1.631294 2.146557 4.713521    38.984497   0.11245671  0.2857060
##   Col.CoV.hrz Col.skew.hrz Col.kurtosis.hrz Col.mean.vrt Col.sd.vrt Col.CoV.vrt
## 1    2.117043     3.926095         22.61682   0.08907545  0.1942661    2.180916
## 2    2.540587     5.473994         47.54446   0.10647940  0.2643248    2.482403
##   Col.skew.vrt Col.kurtosis.vrt Lum.mean.hrz Lum.sd.hrz Lum.CoV.hrz
## 1     3.913759         22.07321    0.4355562  0.8958863    2.056879
## 2     5.224416         45.33121    0.3860825  1.0964629    2.839970
##   Lum.skew.hrz Lum.kurtosis.hrz Lum.mean.vrt Lum.sd.vrt Lum.CoV.vrt
## 1     3.612730         20.01520    0.4563516  0.9904756    2.170422
## 2     7.414604         84.33807    0.3751880  1.0664402    2.842415
##   Lum.skew.vrt Lum.kurtosis.vrt Transform Col.thresh Lum.thresh    GabRat
## 1     3.852029         22.42973      none          0          0 0.1682120
## 2     7.844373         98.05338      none          0          0 0.5332068
```

```r
dim(VCA_LEIA_GabRat_analysis) # 18 rows and 165 columns
```

```
## [1]  18 165
```

6.2 Data cleaning

In this tutorial, we are using a minimal working example, where the data used does not have proper replication (i.e., we have 3 individuals in total). For those considering using linear mixed effects models (LMMs) in R for downstream analyses, we recommend reading Harrison et al. (2018). Please consider the number of data points (n) to parameters (k); where ideally $n/k > 3$ (but see discussion in Harrison et al. 2018).

6.2.1 Omit rows based on values

To demonstrate some data cleaning methods, we have created a smaller data frame `V.L.GbRt_sub_analysis`, subset from `VCA_LEIA_GabRat_analysis` by selecting the columns names. In this data set, there are both `NaN` (not a number) and `Inf` (infinite) entries. Using `summary()` we can see that by retaining `NaN` and `Infs`,

functions like `mean()` and `max()` can not calculated values correctly. To remove `Inf` values, we first convert them to `NA`. To do so, we can use the `is.infinite()` function to detect `Inf` values, however this only works on vectors (i.e., data frame columns), so we can loop over our data frame using a variant of `lapply()`, the `sapply()` function which returns a vector (not list).

```
# Specify column names
ColumnNames <- c(
  "Species", "Ind", "Dist", "ROI", "BSA.BsL.Hrz", "Lum.mean",
  "CAA.Scpl", "Lum.CoV", "BSA.BML", "Col.mean", "GabRat"
)

# Subset data by column names
V.L.GbRt_sub_analysis <- VCA_LEIA_GabRat_analysis[, colnames(VCA_LEIA_GabRat_analysis)
%in% ColumnNames]

dim(V.L.GbRt_sub_analysis) # 18 rows by 11 columns
```

```
## [1] 18 11
```

```
summary(V.L.GbRt_sub_analysis) # Initial summary of the data
```

```
##               Species                                Ind      Dist
##  Aphelodoris varia:12   3_AphelodorisVaria_10_NF_D_RAW:6   30cm:9
##  Aplysia sp       : 6   3_AphelodorisVaria_11_NF_D_RAW:6   50cm:9
##                         4_Aplysiasp_1_NL_N_RAW        :6
##
##
##
##
##                 ROI       CAA.Scpl         BSA.BML        BSA.BsL.Hrz
##  animal           :6   Min.   :0.3013   Min.   :0.0000   Min.   :0.00000
##  animal+background:6   1st Qu.:0.5280   1st Qu.:0.2007   1st Qu.:0.04736
##  background       :6   Median :0.6423   Median :0.2774   Median :0.07755
##                        Mean   :   Inf   Mean   :0.2503   Mean   :0.07439
##                        3rd Qu.:0.8069   3rd Qu.:0.3371   3rd Qu.:0.10934
##                        Max.   :   Inf   Max.   :0.4793   Max.   :0.15655
##                                                          NA's   :4
##     Col.mean         Lum.mean         Lum.CoV          GabRat
##  Min.   :0.1542   Min.   :0.5980   Min.   :1.483   Min.   :0.1624
##  1st Qu.:0.1827   1st Qu.:0.7779   1st Qu.:1.709   1st Qu.:0.2622
##  Median :0.1937   Median :0.8958   Median :1.864   Median :0.3781
##  Mean   :0.2068   Mean   :0.9352   Mean   :1.904   Mean   :0.3521
##  3rd Qu.:0.2291   3rd Qu.:1.0017   3rd Qu.:2.062   3rd Qu.:0.4386
##  Max.   :0.2857   Max.   :1.5512   Max.   :2.450   Max.   :0.5332
##
```

```
# View the data frame
View(V.L.GbRt_sub_analysis)

# Replace infinite values with NA using sapply
V.L.GbRt_sub_analysis[sapply(V.L.GbRt_sub_analysis, is.infinite)] <- NA
```

```r
View(V.L.GbRt_sub_analysis) # Observe change the data frame

# Remove NA/NaNs
V.L.GbRt_sub_analysis <- na.omit(V.L.GbRt_sub_analysis)

View(V.L.GbRt_sub_analysis) # Observe change the data frame

dim(V.L.GbRt_sub_analysis) # 11 rows by 11 columns
```

```
## [1] 11 11
```

```r
summary(V.L.GbRt_sub_analysis) # Inspect column distributions
```

```
##               Species                              Ind      Dist
##  Aphelodoris varia:6   3_AphelodorisVaria_10_NF_D_RAW:2   30cm:7
##  Aplysia sp       :5   3_AphelodorisVaria_11_NF_D_RAW:4   50cm:4
##                        4_Aplysiasp_1_NL_N_RAW        :5
##
##
##
##                 ROI       CAA.Scpl         BSA.BML        BSA.BsL.Hrz
##  animal           :3   Min.   :0.3013   Min.   :0.1332   Min.   :0.04544
##  animal+background:5   1st Qu.:0.4786   1st Qu.:0.2402   1st Qu.:0.06180
##  background       :3   Median :0.5396   Median :0.2888   Median :0.09529
##                        Mean   :0.5261   Mean   :0.2831   Mean   :0.09468
##                        3rd Qu.:0.5966   3rd Qu.:0.3348   3rd Qu.:0.11521
##                        Max.   :0.6832   Max.   :0.3966   Max.   :0.15655
##     Col.mean         Lum.mean         Lum.CoV          GabRat
##  Min.   :0.1542   Min.   :0.5980   Min.   :1.483   Min.   :0.1906
##  1st Qu.:0.1881   1st Qu.:0.8488   1st Qu.:1.769   1st Qu.:0.2678
##  Median :0.2161   Median :0.9803   Median :2.040   Median :0.4105
##  Mean   :0.2179   Mean   :1.0176   Mean   :1.983   Mean   :0.3775
##  3rd Qu.:0.2534   3rd Qu.:1.0842   3rd Qu.:2.104   3rd Qu.:0.4647
##  Max.   :0.2857   Max.   :1.5512   Max.   :2.450   Max.   :0.5332
```

```r
# Inspect the variance for each colour variable
apply(V.L.GbRt_sub_analysis[, 5:11], 2, var)
```

```
##    CAA.Scpl     BSA.BML BSA.BsL.Hrz    Col.mean    Lum.mean     Lum.CoV
## 0.012682464 0.006015432 0.001430767 0.002207957 0.087193128 0.090558647
##      GabRat
## 0.015336849
```

6.2.2 Testing for normality and univaraite outliers

Again, with our minimal example, our estimates do not deviate from normality but here we have provided code to test this. Using a combination of the Shapiro-Wilk test for normality (Part A) and histograms (Part B) we can determine if our data is normal. For the Shapiro-Wilk test, the null hypothesis is that data is normal (i.e. p-value less than 0.05 are not normally distributed).

```r
# Part A:
# Test for normality Shapiro-Wilk test
lapply(V.L.GbRt_sub_analysis[, 5:11], shapiro.test)
```

```
## $CAA.Scpl
##
##  Shapiro-Wilk normality test
##
## data:  X[[i]]
## W = 0.95222, p-value = 0.6724
##
##
## $BSA.BML
##
##  Shapiro-Wilk normality test
##
## data:  X[[i]]
## W = 0.97579, p-value = 0.9383
##
##
## $BSA.BsL.Hrz
##
##  Shapiro-Wilk normality test
##
## data:  X[[i]]
## W = 0.94014, p-value = 0.5222
##
##
## $Col.mean
##
##  Shapiro-Wilk normality test
##
## data:  X[[i]]
## W = 0.92552, p-value = 0.3674
##
##
## $Lum.mean
##
##  Shapiro-Wilk normality test
##
## data:  X[[i]]
## W = 0.90032, p-value = 0.1865
##
##
## $Lum.CoV
##
##  Shapiro-Wilk normality test
##
## data:  X[[i]]
## W = 0.95266, p-value = 0.6781
##
##
## $GabRat
```

```
##
##  Shapiro-Wilk normality test
##
## data:  X[[i]]
## W = 0.90139, p-value = 0.1922
```

```r
# Part B:
# Visualise distribution of the data using histograms
Colnames_subset <- colnames(V.L.GbRt_sub_analysis[, 5:11]) # Select only the colour variables

# Set the plot dimensions to display plots as 3 rows by 3 columns
par(mfrow = c(3, 3))

# Use a for loop to iterate over the column names
for (i in Colnames_subset) {
  # Plot the histogram, using 10 bins, and the colour variable as the main title
  hist(V.L.GbRt_sub_analysis[, i],
    breaks = 10,
    main = i, xlab = ""
  )
}
```

To determine the degree of association between our colour variables, we can use both correlations and scatter plots. As our data is normally distributed we can use Pearson's correlation.

```r
# Correlation panel
panel.cor <- function(x, y) {
  usr <- par("usr")
  on.exit(par(usr))
  par(usr = c(0, 1, 0, 1))

  # Calculate Pearson's correlation
  r <- round(cor(x, y, method = "pearson"), digits = 2)

  # Save as text
  txt <- paste0("R = ", r)

  # Plot the text
  text(0.5, 0.5, txt, cex = 1)
}

my_cols <- c("#00AFBB", "#E7B800", "#FC4E07")

# Scatter plot panel
panel.scat <- function(x, y) {
  # Specify scatter plot points to colour by ROI
  points(x, y,
    col = my_cols[V.L.GbRt_sub_analysis$ROI],
    cex = 1.1, pch = 19
  )
}

pairs(V.L.GbRt_sub_analysis[, 5:11],
  # Set title
```

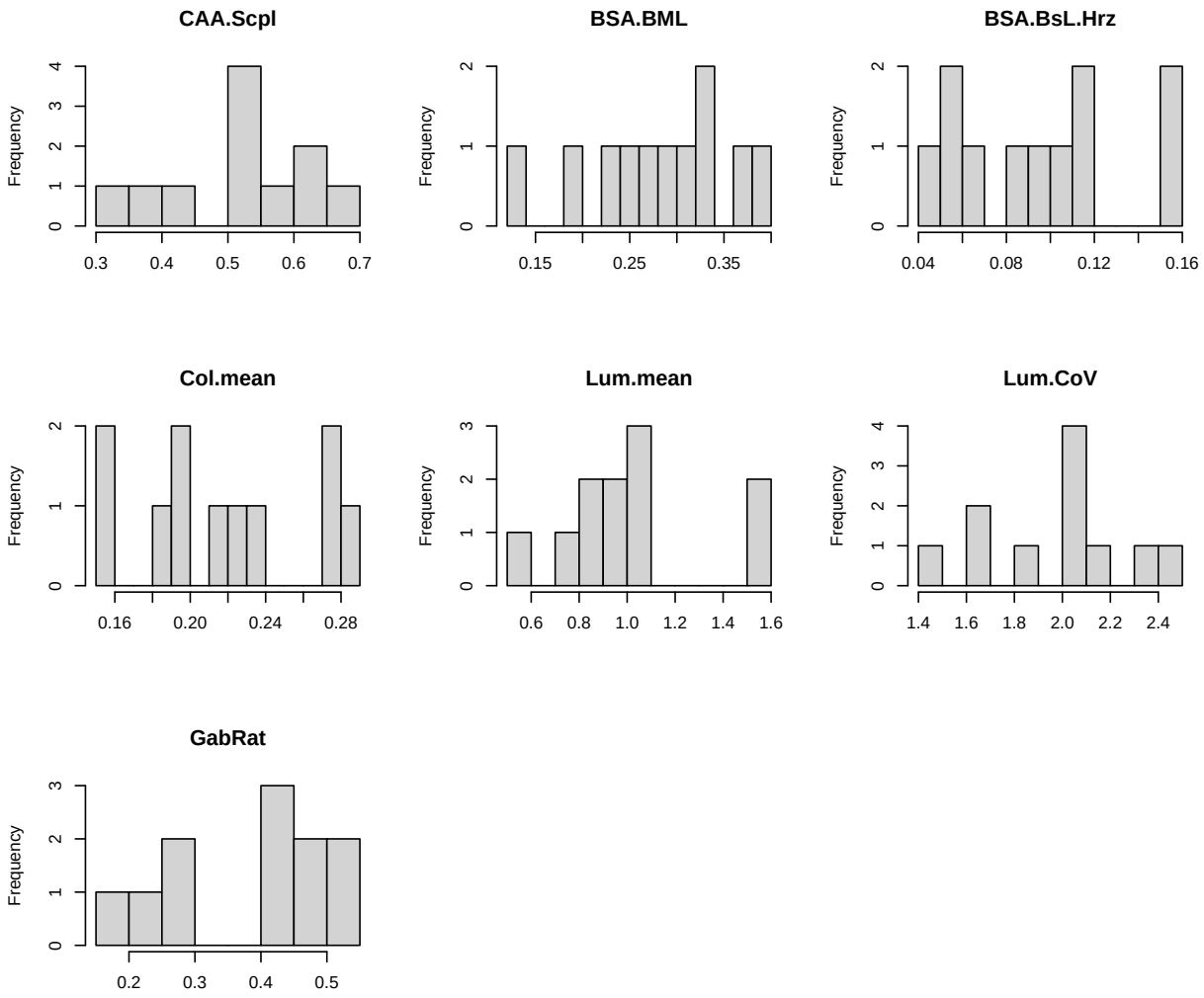

Figure 2: Figure 2. Histograms of seven colour variables.

```r
  main = "Seven colour variables for two viewing distances",

  # Parse in the new panels
  upper.panel = panel.cor,
  lower.panel = panel.scat,

  # Add margin space and reduce gaps between plots
  oma = c(5, 5, 6, 18), gap = 0
)

# Use the data to create a figure legend
Colour_ID <- unique(data.frame(
  as.character(V.L.GbRt_sub_analysis$ROI),
  my_cols[V.L.GbRt_sub_analysis$ROI]
))

# Add figure legend to plot
legend("right", Colour_ID[, 1],
  xpd = TRUE,
  pch = 19, cex = 1.1, bty = "n", col = Colour_ID[, 2]
)
```

From the scatter plots in Figure 3, there doesn't appear to be any outliers in each of the colour variables, however we can confirm this using a boxplot (Part A) and then calculate the z-scores for each estimate (Part B) and look for outliers at greater than 3 standard deviations.

```r
# Part A:
# For the boxplot function, the data has to be in long format
# We need the reshape package for this
library(reshape2)

# Here we 'melt' the data to long format
V.L.GbRt_sub_analysis_lng <- melt(V.L.GbRt_sub_analysis,
  id.vars = c("Species", "Ind", "Dist", "ROI")
)

# Set the plot dimensions to display a single plot, correcting outer margins
par(mfrow = c(1, 1), oma = c(0, 0, 0, 0))

# Inspect boxplots for outliers
boxplot(value ~ variable,
  pch = 16, boxwex = 0.5, xlab = "Colour variable",
  data = V.L.GbRt_sub_analysis_lng
)
```

```r
# Calculate z-scores and look for outliers
apply(V.L.GbRt_sub_analysis[, 5:11], 2, function(x) (x - mean(x)) / sd(x))
```

```
##      CAA.Scpl     BSA.BML BSA.BsL.Hrz    Col.mean    Lum.mean    Lum.CoV
## 2   1.0447329  0.07370189  0.01615151 -0.03748558 -0.87259277  0.5434126
## 3  -1.1591094 -0.04606569 -1.04912193 -0.50269435 -1.42120580 -0.4793016
## 7   1.3944807 -0.67822953 -0.68927858  1.44389073  1.80698261 -1.2187555
```

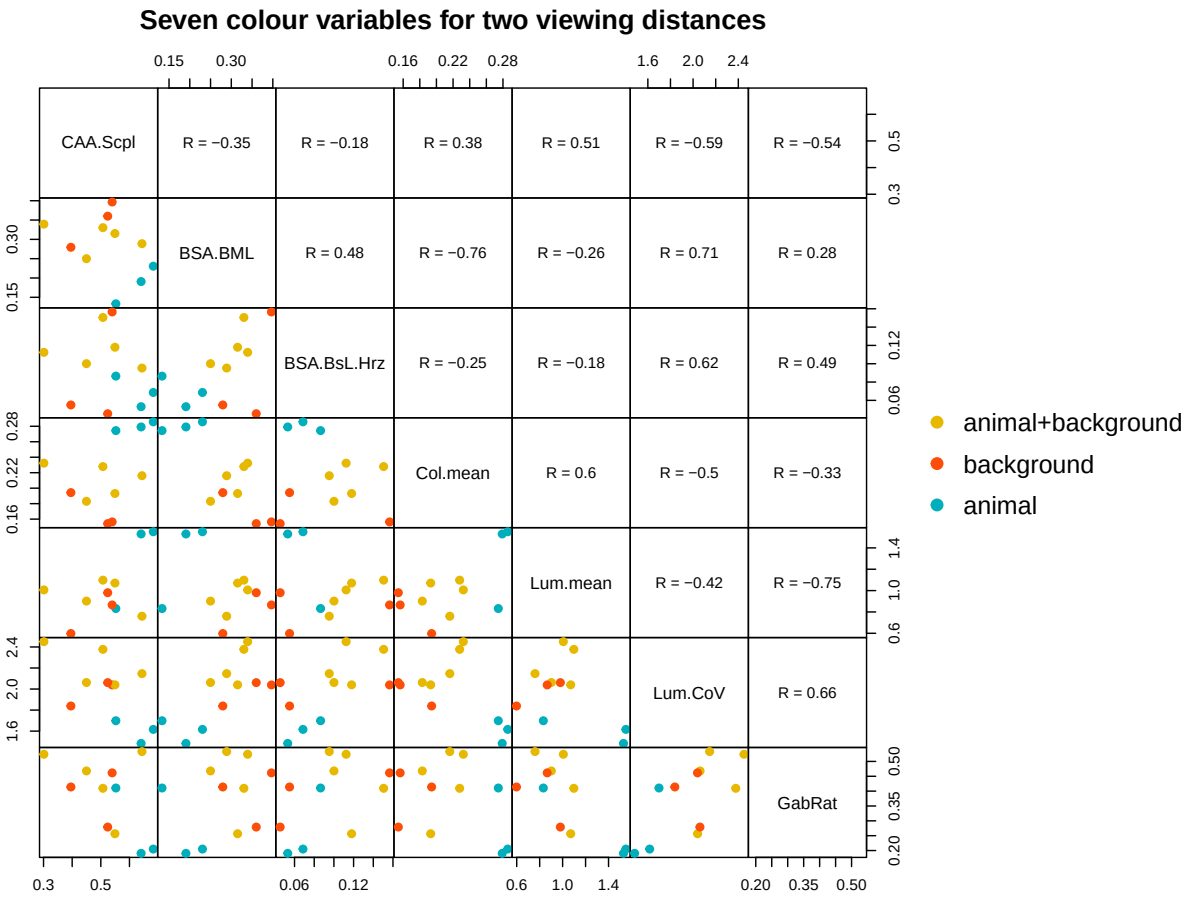

Figure 3: Figure 3. Paired scatter plots of seven colour variables. Lower off-diagonal are the paired scatterplots where coloured points represent regions of interest (animal+background is yellow, bacground is red and animal is blue). Upper off-diagonal represents Pearson's correlation between paired colur varaibles.

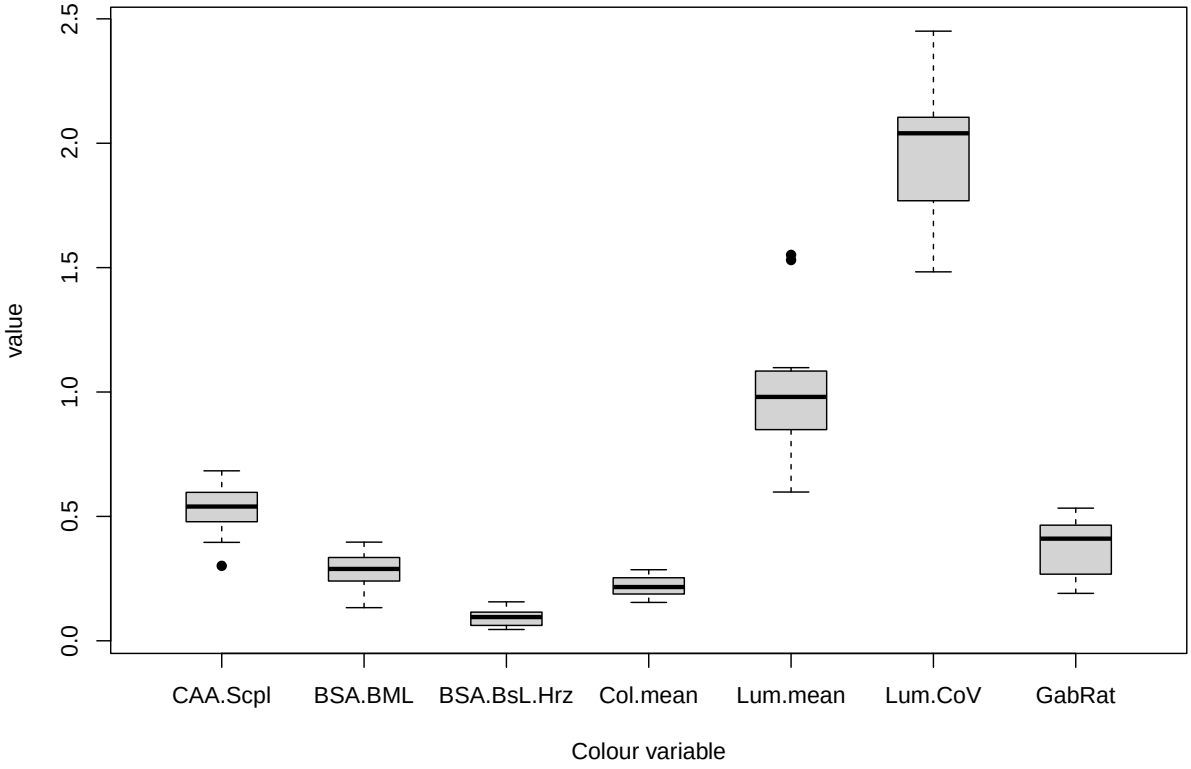

Figure 4: Figure 4. Boxplots of the seven colour variables.

```
## 8  -1.9965592  0.72545366  0.46902751  0.30774049 -0.03805163  1.5533929
## 10  1.0175188 -1.19110556 -1.09924489  1.30323751  1.73650453 -1.6615115
## 11 -0.1664538  0.60802358  1.47594374  0.21417235  0.27164927  1.3104953
## 13  0.2343956 -1.93327402 -0.21669972  1.20297119 -0.62888125 -0.9445464
## 14 -0.6773729 -0.42764791  0.14299748 -0.74177818 -0.39271849  0.2623784
## 15  0.1198638  1.46361125  1.63571008 -1.30800444 -0.51469509  0.1861873
## 17  0.2065297  0.41346282  0.61625213 -0.52704120  0.17954864  0.1893369
## 18 -0.0180262  0.99206954 -1.30173733 -1.35500854 -0.12654002  0.2589115
##        GabRat
## 2   1.2575825
## 3   0.2949464
## 7  -1.3922980
## 8   1.1835279
## 10 -1.5085298
## 11  0.2588835
## 13  0.2669827
## 14  0.7306342
## 15  0.6787438
## 17 -0.9760417
## 18 -0.7944315
```

Although in Figure 4, there are two points in Lum.mean that exceed 1.5 times the interquartile range (the outer whiskers of the boxplot), none of the points exceed 3 standard deviations.

6.2.3 Standardising data

When implementing multivariate models, it is best to standardize data to facilitate model convergence. Here we variance standardised (mean = 0, standard deviation = 1) each colour pattern variable across both viewing distances and ROIs using the `scale()` function.

```r
# Inspect scaled values and column means
scale(V.L.GbRt_sub_analysis[, 5:11], center = TRUE, scale = TRUE)
```

```
##      CAA.Scpl     BSA.BML BSA.BsL.Hrz    Col.mean    Lum.mean    Lum.CoV
## 2   1.0447329  0.07370189  0.01615151 -0.03748558 -0.87259277  0.5434126
## 3  -1.1591094 -0.04606569 -1.04912193 -0.50269435 -1.42120580 -0.4793016
## 7   1.3944807 -0.67822953 -0.68927858  1.44389073  1.80698261 -1.2187555
## 8  -1.9965592  0.72545366  0.46902751  0.30774049 -0.03805163  1.5533929
## 10  1.0175188 -1.19110556 -1.09924489  1.30323751  1.73650453 -1.6615115
## 11 -0.1664538  0.60802358  1.47594374  0.21417235  0.27164927  1.3104953
## 13  0.2343956 -1.93327402 -0.21669972  1.20297119 -0.62888125 -0.9445464
## 14 -0.6773729 -0.42764791  0.14299748 -0.74177818 -0.39271849  0.2623784
## 15  0.1198638  1.46361125  1.63571008 -1.30800444 -0.51469509  0.1861873
## 17  0.2065297  0.41346282  0.61625213 -0.52704120  0.17954864  0.1893369
## 18 -0.0180262  0.99206954 -1.30173733 -1.35500854 -0.12654002  0.2589115
##        GabRat
## 2   1.2575825
## 3   0.2949464
## 7  -1.3922980
## 8   1.1835279
## 10 -1.5085298
## 11  0.2588835
## 13  0.2669827
```

```
## 14  0.7306342
## 15  0.6787438
## 17 -0.9760417
## 18 -0.7944315
## attr(,"scaled:center")
##    CAA.Scpl     BSA.BML BSA.BsL.Hrz    Col.mean    Lum.mean     Lum.CoV
##   0.5261440   0.2831279   0.0946824   0.2178738   1.0176215   1.9830285
##      GabRat
##   0.3774653
## attr(,"scaled:scale")
##    CAA.Scpl     BSA.BML BSA.BsL.Hrz    Col.mean    Lum.mean     Lum.CoV
##  0.11261644  0.07755922  0.03782548  0.04698890  0.29528483  0.30092964
##      GabRat
##  0.12384203
```

```r
# Update the V.L.GbRt_sub_analysis data frame with the standardised values
V.L.GbRt_sub_analysis[, 5:11] <- scale(V.L.GbRt_sub_analysis[, 5:11],
  center = TRUE, scale = TRUE
)

# Check standardisation
summary(V.L.GbRt_sub_analysis) # Mean values are zero
```

```
##               Species                                Ind      Dist
##  Aphelodoris varia:6   3_AphelodorisVaria_10_NF_D_RAW:2   30cm:7
##  Aplysia sp       :5   3_AphelodorisVaria_11_NF_D_RAW:4   50cm:4
##                        4_Aplysiasp_1_NL_N_RAW        :5
##
##
##
##                 ROI       CAA.Scpl          BSA.BML         BSA.BsL.Hrz
##  animal           :3   Min.   :-1.9966   Min.   :-1.9333   Min.   :-1.30174
##  animal+background:5   1st Qu.:-0.4219   1st Qu.:-0.5529   1st Qu.:-0.86920
##  background       :3   Median : 0.1199   Median : 0.0737   Median : 0.01615
##                        Mean   : 0.0000   Mean   : 0.0000   Mean   : 0.00000
##                        3rd Qu.: 0.6260   3rd Qu.: 0.6667   3rd Qu.: 0.54264
##                        Max.   : 1.3945   Max.   : 1.4636   Max.   : 1.63571
##     Col.mean           Lum.mean          Lum.CoV            GabRat
##  Min.   :-1.35501   Min.   :-1.4212   Min.   :-1.6615   Min.   :-1.5085
##  1st Qu.:-0.63441   1st Qu.:-0.5718   1st Qu.:-0.7119   1st Qu.:-0.8852
##  Median :-0.03749   Median :-0.1265   Median : 0.1893   Median : 0.2670
##  Mean   : 0.00000   Mean   : 0.0000   Mean   : 0.0000   Mean   : 0.0000
##  3rd Qu.: 0.75536   3rd Qu.: 0.2256   3rd Qu.: 0.4029   3rd Qu.: 0.7047
##  Max.   : 1.44389   Max.   : 1.8070   Max.   : 1.5534   Max.   : 1.2576
```

```r
apply(V.L.GbRt_sub_analysis[, 5:11], 2, var) # Confirm variances are 1
```

```
##    CAA.Scpl     BSA.BML BSA.BsL.Hrz    Col.mean    Lum.mean     Lum.CoV
##           1           1           1           1           1           1
##      GabRat
##           1
```

6.2.4 Multivaraite outliers

For users who will be investigating colours multivariately, we would recommend using Mahalanobis distance to determine if there are any outliers in multivariate space. To determine p-values for Mahalanobis distance, you can use the Chi-Square statistic with k-1 degrees of freedom, where k is the number of variables (here k = 7), and p-values less than 0.05 outliers.

```
# Using R's Mahalanobis function, add a new column to your data
V.L.GbRt_sub_analysis$mahalnobis <- mahalanobis(
  V.L.GbRt_sub_analysis[, 5:11],

  # Get a vector of column means
  colMeans(V.L.GbRt_sub_analysis[, 5:11]),

  # Get a covaraince matrix
  cov(V.L.GbRt_sub_analysis[, 5:11])
)

# Calcualte p-values using k-1 df = 6
V.L.GbRt_sub_analysis$pvalue <- pchisq(V.L.GbRt_sub_analysis$mahalnobis,
  df = 6, lower.tail = FALSE
)

# Inspect Mahalnobis distances and p-values
V.L.GbRt_sub_analysis[, colnames(V.L.GbRt_sub_analysis) %in% c("mahalnobis", "pvalue")]
```

```
##    mahalnobis    pvalue
## 2    7.941035 0.2424552
## 3    6.662854 0.3531541
## 7    4.371090 0.6265923
## 8    7.589055 0.2697817
## 10   4.524537 0.6060684
## 11   4.608575 0.5949021
## 13   6.112840 0.4106692
## 14   8.005528 0.2376986
## 15   8.490756 0.2043073
## 17   5.134891 0.5266326
## 18   6.558839 0.3635764
```

7 Export files

Depending on what image analysis sets you are using, we have provided multiple ways to save the image analysis data files as csvs in the folder 'Output'. First, we use `dir.create()` to create the folder 'Output' in the Test Data folder, and save this new folder location as `Out_path`. To save a single file, you can use `write.csv()`, where we have used the argument `row.names = FALSE` to make sure that the row numbers in the dataframe aren't included in the csv file.

To save mutliple files, we utilise the fact that our naming convention in this R tutorial specifies image analysis data files with `_analysis`. Using the `ls()` function with the pattern argument set to `_analysis`, we can create a list of all our image analysis dataframes. Then, using a for loop, we iterate over each dataframe, rename it using the current date, and then output the csv file. Finally, we use `list.files()` to check that the files have been created.

```r
dir.create(paste(c(loc, "Output"), collapse = "/")) # Create the Output directory

Out_path <- paste(c(loc, "Output"), collapse = "/") # Save Output directory path

# For a single file:
write.csv(GabRat_Res_analysis,
  row.names = FALSE, # Remove row names
  file = paste(c(Out_path, "GabRat_Res_analysis.csv"),
    collapse = "/"
  )
) # Save to the folder Output in Test Data

# For multiple files:
savefile_list <- ls(pattern = "analysis") # Save all objects with "analysis" in the name

savefile_list # Inspect object names
```

```
##  [1] "Cluster_analysis"          "Cluster_Particle_analysis"
##  [3] "GabRat_Res_analysis"       "IndParticle_analysis"
##  [5] "LEIA_Res_analysis"         "Particle_analysis"
##  [7] "V.L.GbRt_sub_analysis"     "V.L.GbRt_sub_analysis_lng"
##  [9] "VCA_analysis"              "VCA_LEIA_GabRat_analysis"
```

```r
# Iterate over each data frame in savefile_list
for (i in savefile_list) {
  # Create file name and add the .csv extension
  # Add date of creation using format(Sys.time(), "_%d_%b_%Y")
  # Remove "_analysis" using gsub and add "_data" and date created
  fname <- paste0(gsub("_analysis", "", i), "_data", format(Sys.time(), "_%d_%b_%Y"), ".csv")

  # Save to the folder Output in Test Data
  write.csv(GabRat_Res_analysis,
    row.names = FALSE, # Remove row names
    file = paste(c(Out_path, fname), collapse = "/")
  )
}

# Check that your files have saved
list.files(Out_path)
```

```
##  [1] "Cluster_data_27_Oct_2023.csv"
##  [2] "Cluster_Particle_data_27_Oct_2023.csv"
##  [3] "GabRat_Res_analysis.csv"
##  [4] "GabRat_Res_data_27_Oct_2023.csv"
##  [5] "IndParticle_data_27_Oct_2023.csv"
##  [6] "LEIA_Res_data_27_Oct_2023.csv"
##  [7] "Particle_data_27_Oct_2023.csv"
##  [8] "V.L.GbRt_sub_data_27_Oct_2023.csv"
##  [9] "V.L.GbRt_sub_lng_data_27_Oct_2023.csv"
## [10] "VCA_data_27_Oct_2023.csv"
## [11] "VCA_LEIA_GabRat_data_27_Oct_2023.csv"
```

8 Fast import for large data sets using data.table (optional - advanced)

For very large data sets we would recommend importing data using the `data.table` package (Dowle and Srinivasan 2023). The `data.table` format can be used in parallel and is optimised for fast `.csv` reading/writing. For further information and cheatsheets, visit the `data.table` GitHub page (https://github.com/Rdatatable/data.table). The *data.table* has the following structure:

```
FROM[WHERE, SELECT, GROUP BY]
DT  [i,     j,      by]
```

Here we will provide a quick demonstration by importing the LEIA QCPA batch script output. First, we convert our `Animal_Info_ROI` into `data.table` format using the function `setDT`. Then we will input our data using the dedicated `data.table` read function `fread`. Finally, we can output large data.tables fast using the `fwrite` function.

```r
library(data.table)
```

```
##
## Attaching package: 'data.table'

## The following objects are masked from 'package:reshape2':
##
##     dcast, melt
```

```r
# Convert Animal_Info_ROI into data.table format
Animal_Info_ROI_DT <- setDT(Animal_Info_ROI)

# From Animal_Info_ROI we can paste the column information together
# and then read in data row by row, specifying 'by=1:NROW(Animal_Info_ROI)'
LEIA_analysis_DT <- Animal_Info_ROI[, fread(paste(loc, Species, Ind, Dist, "LEIA", ROI,
  "_Local Edge Intensity Analysis.csv",
  sep = "/"
)),
by = 1:NROW(Animal_Info_ROI)
]

# Remove unwanted columns, and automatically assign using ':='
LEIA_analysis_DT[, c("NROW", "V1", "Image", "Transform") := NULL]

# Add the Animal_Info_ROI information
LEIA_analysis_DT <- cbind(Animal_Info_ROI, LEIA_analysis_DT)

# Inspect the data.table
head(LEIA_analysis_DT)
```

```
##              Species                            Ind Dist    ROI  Col mean
## 1: Aphelodoris varia 3_AphelodorisVaria_10_NF_D_RAW 30cm animal 0.1736449
## 2: Aphelodoris varia 3_AphelodorisVaria_10_NF_D_RAW 50cm animal 0.1663600
## 3: Aphelodoris varia 3_AphelodorisVaria_11_NF_D_RAW 30cm animal 0.2857206
## 4: Aphelodoris varia 3_AphelodorisVaria_11_NF_D_RAW 50cm animal 0.2791115
```

```
## 5:        Aplysia sp         4_Aplysiasp_1_NL_N_RAW 30cm animal 0.2744001
## 6:        Aplysia sp         4_Aplysiasp_1_NL_N_RAW 50cm animal 0.1879060
##       Col sd  Col CoV  Col skew Col kurtosis  Lum mean   Lum sd  Lum CoV
## 1: 0.2851837 1.642339 2.3513009     9.700354 0.8928419 1.431054 1.602808
## 2: 0.2695337 1.620183 2.2863201     9.671851 0.8987935 1.380676 1.536144
## 3: 0.4620673 1.617200 2.1109772     8.397064 1.5511960 2.507150 1.616269
## 4: 0.4172110 1.494782 1.6695742     5.947408 1.5303849 2.269607 1.483030
## 5: 0.5186319 1.890057 0.9863623     3.941409 0.8319224 1.413259 1.698786
## 6: 0.3406233 1.812732 0.8951362     3.821529 0.9877660 1.719976 1.741279
##     Lum skew Lum kurtosis Col mean hrz Col sd hrz Col CoV hrz Col skew hrz
## 1: 2.2559735     9.558368   0.08770301  0.1856711    2.117043     3.926095
## 2: 2.0603133     8.282279   0.08205933  0.1756506    2.140532     3.887552
## 3: 2.5463566    12.453982   0.15267951  0.3221450    2.109943     3.683355
## 4: 2.0211324     8.743586   0.14939446  0.2985774    1.998584     3.244117
## 5: 0.4658983     2.619744   0.13644481  0.3012997    2.208217     2.138189
## 6: 0.6844910     3.147853   0.09580569  0.1994786    2.082117     2.279845
##    Col kurtosis hrz Col mean vrt Col sd vrt Col CoV vrt Col skew vrt
## 1:         22.61682   0.08907545  0.1942661    2.180916     3.913759
## 2:         22.67175   0.08694090  0.1873940    2.155418     3.881844
## 3:         20.80604   0.13751606  0.2817287    2.048696     3.515951
## 4:         16.06562   0.13517600  0.2483763    1.837429     2.907237
## 5:         11.13582   0.13592239  0.3143010    2.312356     2.280570
## 6:         13.36001   0.09162023  0.2063477    2.252207     2.295968
##    Col kurtosis vrt Lum mean hrz Lum sd hrz Lum CoV hrz Lum skew hrz
## 1:         22.07321    0.4355562  0.8958863    2.056879     3.612730
## 2:         22.90526    0.4406635  0.8761384    1.988225     3.493638
## 3:         18.79095    0.8198832  1.7714397    2.160600     4.465145
## 4:         14.14649    0.8071571  1.6670532    2.065339     3.871669
## 5:         10.90464    0.4030806  0.7863049    1.950739     1.070465
## 6:         11.87647    0.4673162  0.9458438    2.023991     1.530686
##    Lum kurtosis hrz Lum mean vrt Lum sd vrt Lum CoV vrt Lum skew vrt
## 1:        20.015200    0.4563516  0.9904756    2.170422     3.852029
## 2:        19.455513    0.4573795  0.9806892    2.144148     3.690217
## 3:        31.933329    0.7301569  1.4871986    2.036821     3.836817
## 4:        23.224363    0.7215794  1.3089950    1.814069     3.191745
## 5:         4.311739    0.3949017  0.8047263    2.037789     1.771279
## 6:         7.331437    0.4531541  1.0669676    2.354536     2.801707
##    Lum kurtosis vrt Col thresh Lum thresh
## 1:         22.42973          0          0
## 2:         20.01364          0          0
## 3:         23.39465          0          0
## 4:         18.53788          0          0
## 5:         10.55201          0          0
## 6:         15.67831          0          0
```

```r
# Create save filename
fname <- paste0(
  Out_path, "/", gsub("_analysis", "", "LEIA_analysis_DT"),
  "_data", format(Sys.time(), "_%d_%b_%Y"), ".csv"
)
# Save output
fwrite(LEIA_analysis_DT, file = fname, row.names = FALSE)
```

References

Dowle, Matt, and Arun Srinivasan. 2023. *Data.table: Extension of 'Data.frame'*. https://CRAN.R-project.org/package=data.table.

Harrison, Xavier A, Lynda Donaldson, Maria Eugenia Correa-Cano, Julian Evans, David N Fisher, Cecily ED Goodwin, Beth S Robinson, David J Hodgson, and Richard Inger. 2018. "A Brief Introduction to Mixed Effects Modelling and Multi-Model Inference in Ecology." *PeerJ* 6: e4794.

van den Berg, Cedric P, Jolyon Troscianko, John A Endler, N Justin Marshall, and Karen L Cheney. 2020. "Quantitative Colour Pattern Analysis (QCPA): A Comprehensive Framework for the Analysis of Colour Patterns in Nature." *Methods in Ecology and Evolution* 11 (2): 316–32.